Daneben Projekt: Hardware & Enclosure 3D Design

Technical representation of the custom IoT enclosure, PCB structure, and IP67 sealing mechanisms.

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1. Exterior Design & Environmental Protection

The external housing is designed to withstand harsh outdoor conditions while providing optimal angles for both solar harvesting and sensor field-of-view. The enclosure utilizes a dark grey, UV-resistant polycarbonate material to prevent long-term sun degradation.

> **Key Exterior Features:**
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> - **Sloped Roof (20-Degree):** Prevents water pooling, sheds snow, and angles the flush-mounted monocrystalline solar panel optimally towards the sun.
> - **Drip Edge:** A protruding overhang above the main casing seam ensures that rain drips completely off the body rather than running down into the sealing gaskets.
> - **Recessed Camera/Sensor Window:** Positioned at the bottom facing downwards at a 45-degree angle. The recess acts as a lens hood, preventing rain droplets from blurring the lens and blocking direct sun glare.



Figure 1: Exterior 3D concept showing the sloped roof, flush solar panel, drip edge, and recessed bottom sensors.

2. Internal Layout & Exploded View

The internal architecture is strictly modular. The goal of this layout is to keep the delicate electronics completely isolated from the user, even during routine maintenance (like battery swaps).

Internal Components & Structure:

- **Inverted T-Shape PCB:** The custom green printed circuit board is shaped to fit the housing perfectly. Power management (solar MPPT) sits at the top, the ESP32-S3 MCU in the middle, and the camera/ToF sensors along the bottom edge.
- **Silicone O-Ring Seals:** Bright red, continuous silicone gaskets are seated in a dedicated groove between the top and bottom casing halves. When compressed by stainless steel screws, this guarantees an **IP67 rating** (dust-tight and protected against water immersion).
- **Isolated Battery Cartridge System:** Notice the cylindrical 18650 battery sliding in from the bottom. It sits inside its own sealed tubular chamber within the enclosure. The battery cap has its own O-ring. *This is a critical design choice:* municipal workers can unscrew the cap and swap the battery without ever exposing the main PCB to humid outside air or rain.

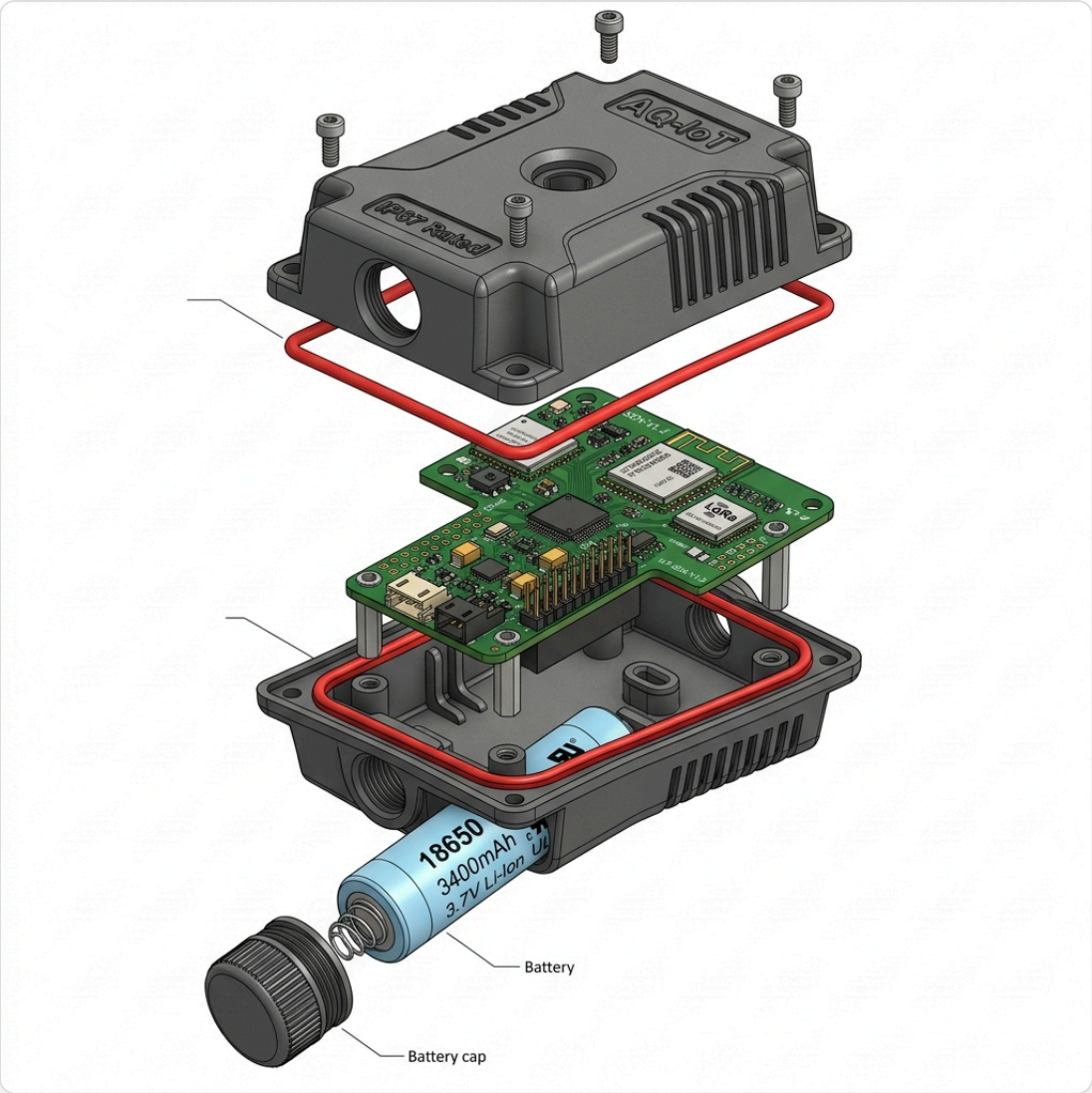


Figure 2: Exploded 3D CAD representation showing the silicone O-rings, custom PCB layout, and the isolated bottom-loading battery cartridge.

3. Usability & Installation Integration

By combining the self-sufficient power system (Solar + 18650 LiFePO4 cells) with a completely sealed form factor, the device is a true "plug and play" node. The backplate features integrated VESA-style slot rails, allowing it to be securely strapped to standard 120L or 240L wheelie bins using heavy-duty steel bands or industrial zip-ties in under two minutes, with zero external wiring required.